

Lab 6

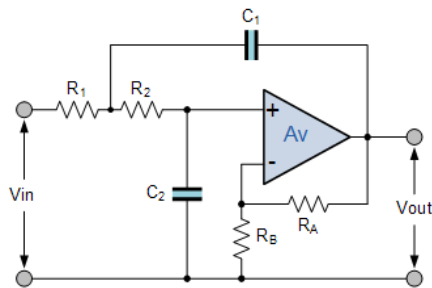
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Introduction:

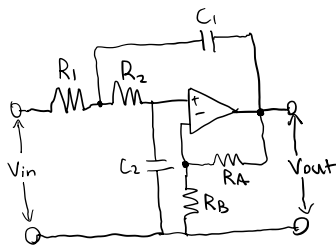
In this lab, we were tasked with building a circuit that takes a single audio frequency and will produce three different outputs. These outputs will consist of three outputs: a low-frequency channel, a mid-frequency channel, and a high-frequency channel. The low-frequency channel will have frequency input of below 300 Hz, the mid-frequency channel will have frequency input of 300 Hz to 5000 Hz, and the high-frequency channel will have a frequency input of above 5000 Hz. These channels will work as a three-way speaker system for an audio signal.

Part 1:

Circuit diagram of second-order low pass filter:



Calculations:



$$f_c = \frac{1}{2\pi RC} \quad R = \frac{1}{2\pi(700)(0.047 \times 10^{-6})}$$

$$R = 16,931.37 \quad R = 17K$$

$$C = 0.047 nF$$

$$G = 1 + \frac{R_A}{R_B}$$

$$5 = 1 + \frac{R_A}{R_B}$$

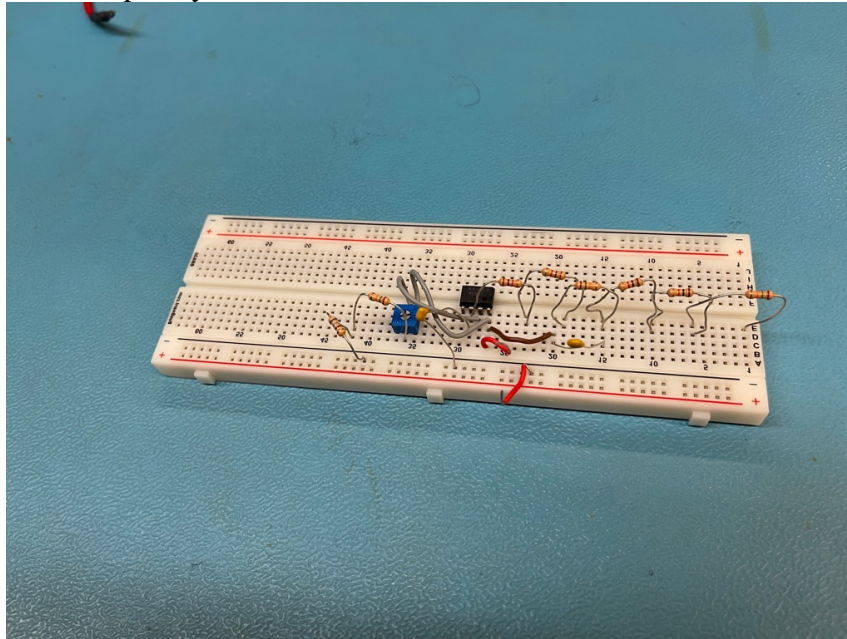
$$4 = \frac{R_A}{R_B} = R_B(4) = R_A$$

$$R_B = \frac{R_A}{4}$$

$$G = 1 + \frac{10k}{2.5k}$$

These are our calculations for the second order low-pass filter. We found our values for C and R using the formula for finding high corner frequency. The value that we found is 17k for R1 and R2, 0.047 μF for C1 and C2, 10k for RA, and 2.5k for RB. Creating this circuit with these values and using an input frequency of 200 Hz should give us a gain of 5. Using a 10k potentiometer for RA we should be able to adjust the gain from 1 to 5 which is our desired result.

Low Frequency Channel Circuit:

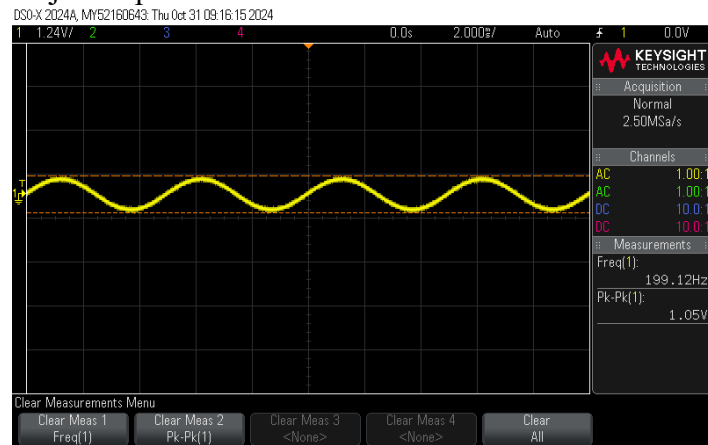


This is our low frequency channel circuit; we built it using a second-order low pass filter. We also implemented a potentiometer in our design, so we can manipulate the gain from 1 to 5. It also has a high corner frequency of 200 Hz. To replicate the value for R1 and R2, I put three resistors in series one 15k and two 1k. One issue we ran into was that our circuit would not output the correct output voltage. In our original design we used a 10k potentiometer and a 2.2k resistor. We ended up getting the wrong output voltage values and it did not give us the correct adjustable gain. We ultimately decided to use a 100k potentiometer and two resistors in series a 15k and a 680 for RB. Surprisingly, adding this change helped our circuit work perfectly. After using these new components, we were finally able to get our correct adjustable gain of 1 to 5.

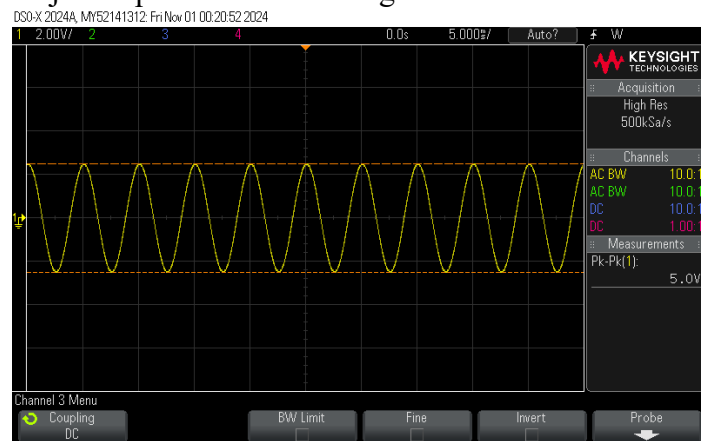
Oscilloscope Waveforms

Using an input voltage of sine wave at 1 Vp-p and a frequency of 200 Hz we get the following output waveforms.

Adjusted potentiometer to lowest value:



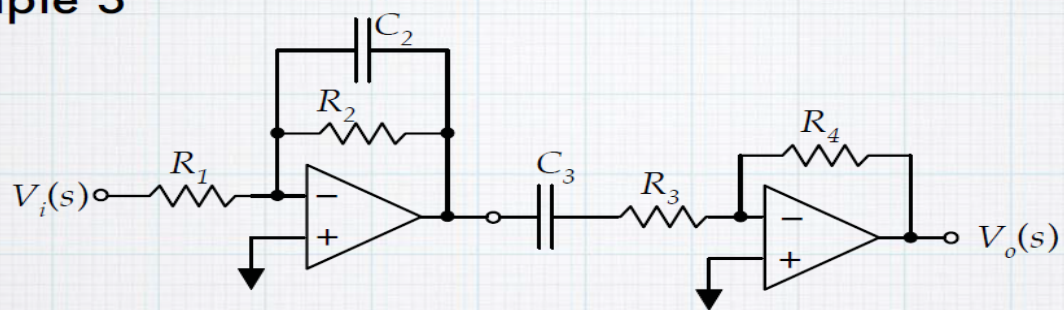
Adjusted potentiometer to highest value:



Part 2:

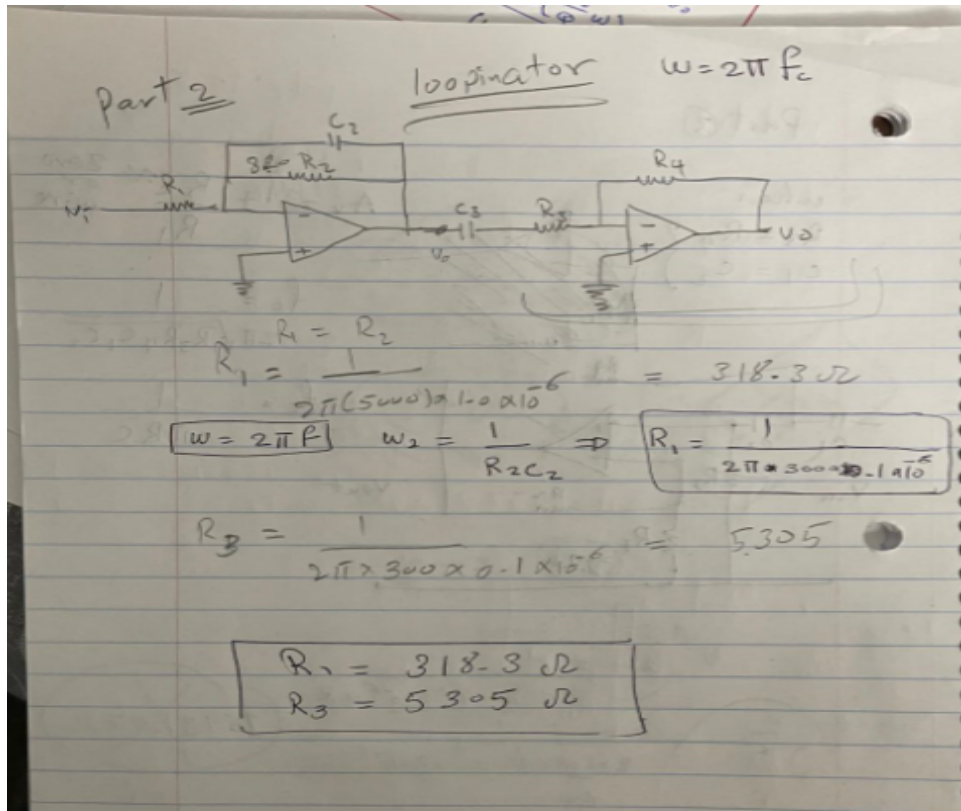
In this section of the lab, we construct a second-order bandpass filter by combining a low-pass filter with a high-pass filter.

Example 3

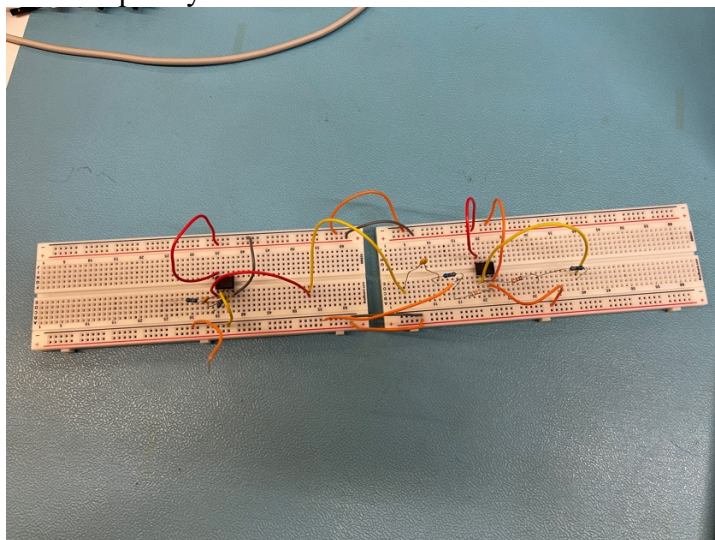


Calculations:

In this section, a low corner frequency of 300 Hz is employed, along with a capacitor value of 0.01 microfarads. This configuration results in a resistor value of 318 ohms, and for simplicity, we set R_1 equal to R_2 . On the right side, in the high-pass filter, R_3 is equal to R_4 , with a capacitor of 0.01 microfarads and a high-pass frequency of 5000 Hz, yielding a resistor value of 5305 ohms.



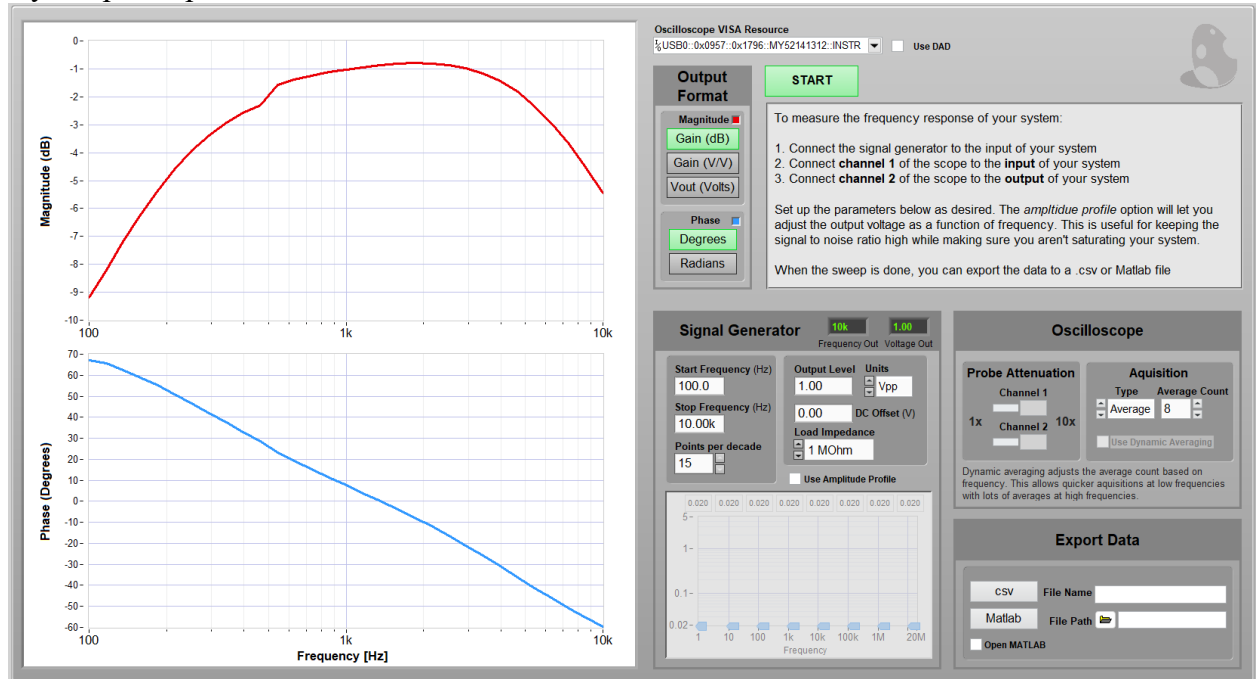
Mid Frequency Channel Circuit :



In our circuit design, we used for resistor values of 330 ohms for the low pass filter, instead from the calculated value of 318 ohms. Additionally, for the high pass filter, we utilized resistor values of 5300 ohms by combining a 5k ohm resistor with a 300 ohm resistor, rather than using the calculated value of 5305 ohms.

We will use loopinator software to create the frequency response plots. Loopinator is a software used to capture the frequency response of a physical circuit. It does this by capturing the output at increasing frequencies and plotting these points on a graph creating a line graph.

Frequency Response plot:



We utilize the Loopinator device to produce 15 frequency samples ranging from 100 Hz to 10 kHz, and the resulting graph is illustrated in the diagram above. The behavior of this bandpass filter indicate that it allows frequencies near 1225 Hz to pass through with minimal attenuation, while significantly attenuating frequencies below 300 Hz and above 5000 Hz. Center of bandwidth frequency = 1224.744 Hz.

$$f_c = \sqrt{f_{low} * f_{high}}$$

$$f_c = \sqrt{300 * 5000} = 1224.744$$

Figure below shows low pass filter behavior in loopinator:

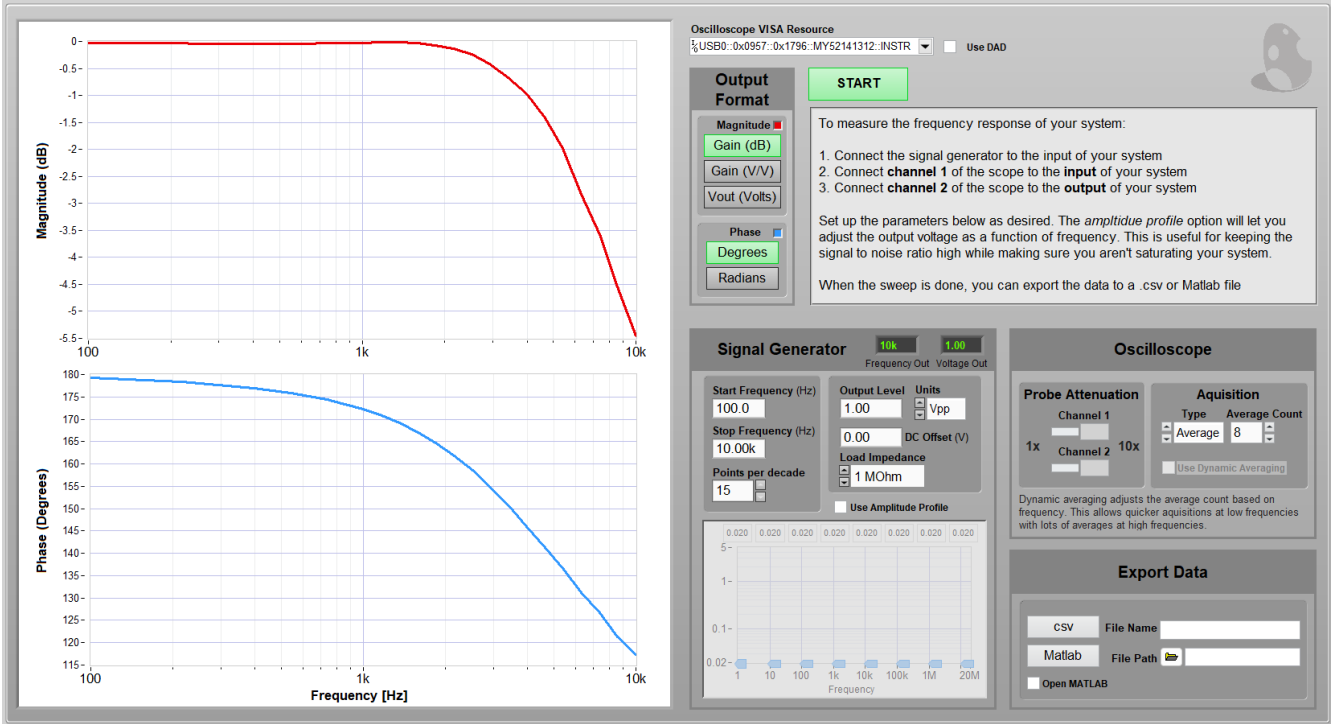
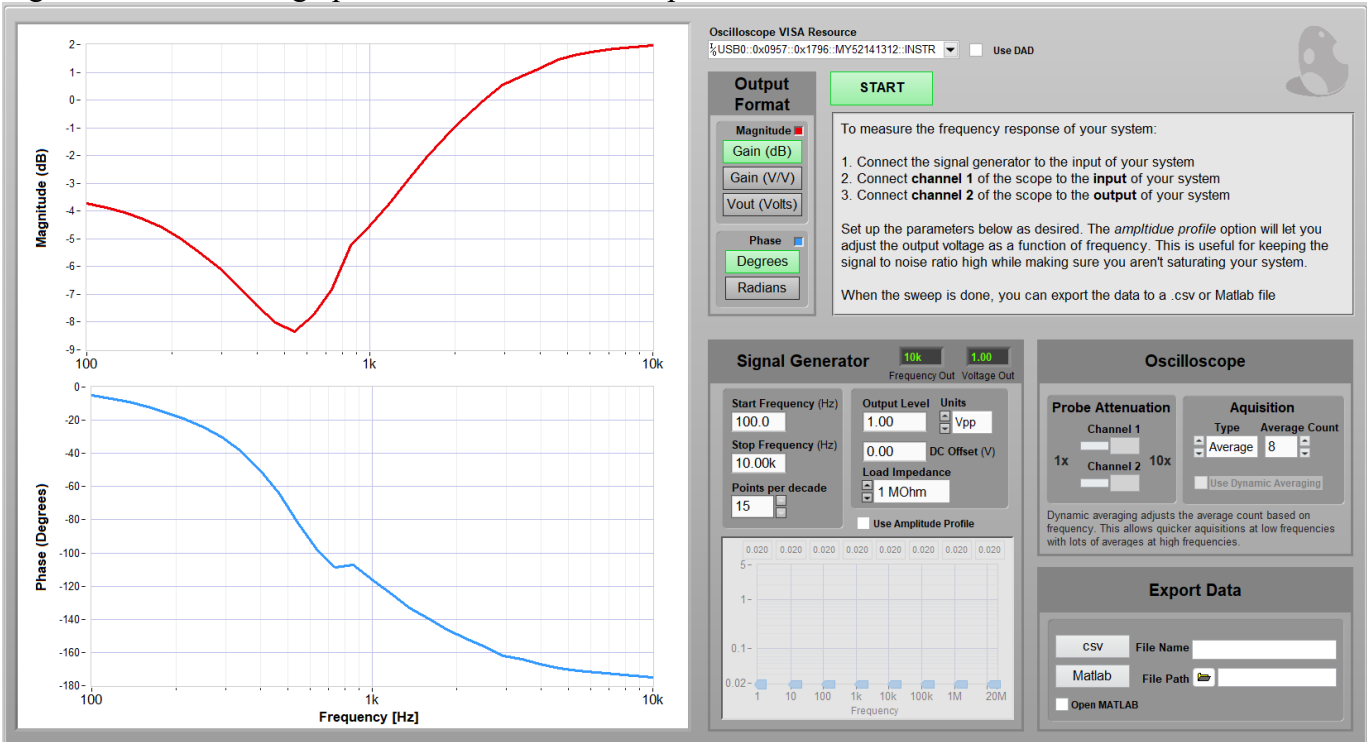
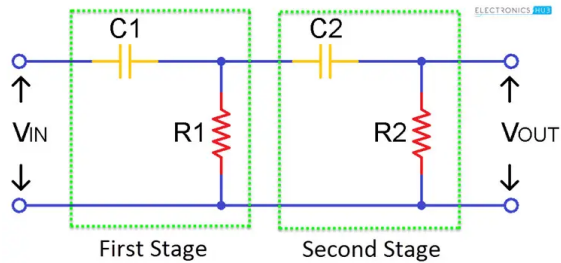


Figure below shows high pass filter behavior in loopinator:

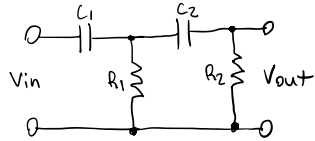


Part 3:

Circuit diagram of second-order passive high pass filter



Calculations:



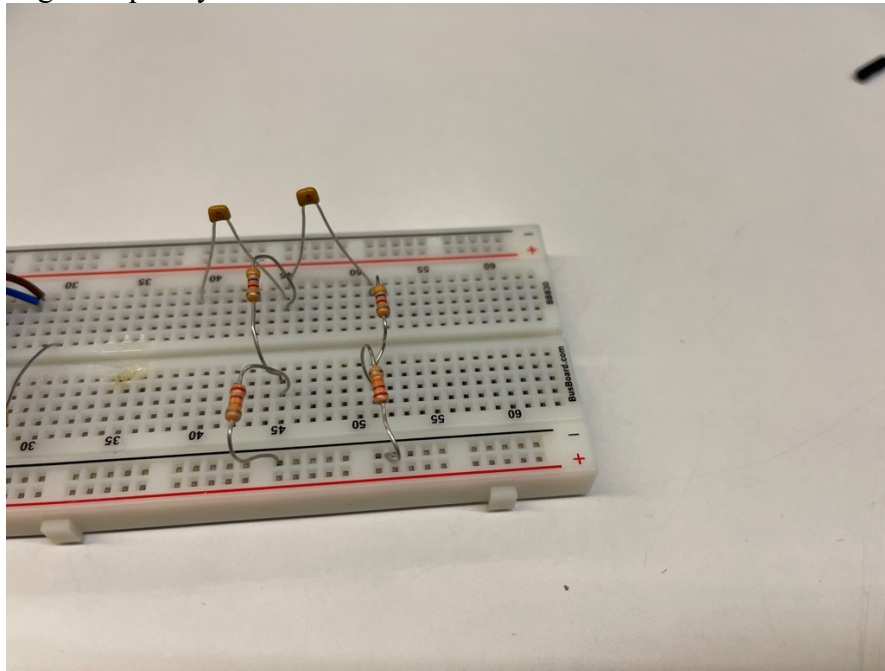
$$f_c = \frac{1}{2\pi RC} \quad C = 0.001 \mu F$$
$$f_c = 6000 \text{ Hz}$$

$$R = \frac{1}{2\pi(6000)(0.001 \times 10^{-6})}$$
$$R = 26525.82 \quad R = 27 \text{ k}$$

$$\text{Gain} = \frac{V_o}{V_{in}} \quad \text{Gain}_{dB} = 20 \log(V_o/V_{in})$$

These are our calculations for the second order passive high-pass filter. We found our values for C and R using the formula for finding low corner frequency. The value we found for R1 and R2 is 27k and 0.001 μF for C1 and C2. Creating this circuit with these values we should get a circuit that has gain of less than 1 and a high corner frequency of 10 kHz.

High Frequency Channel Circuit:

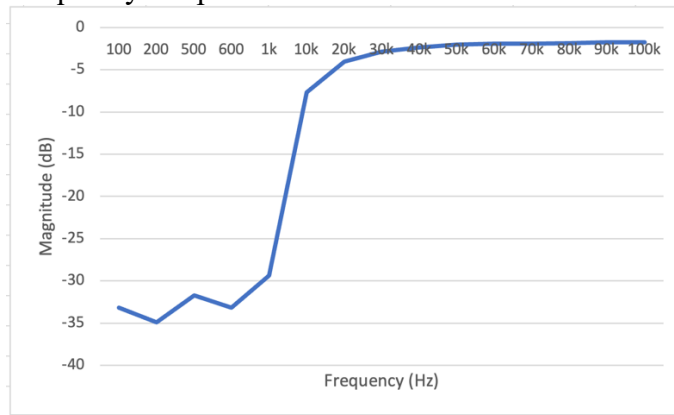


This is the high frequency channel circuit; we built using a second order passive high-pass filter. This circuit has a gain of less than one and a low corner frequency of 6000 Hz. We use two resistors in series to replicate the value of R_1 and R_2 . We have a 4.7k resistor and 22k resistor in series. We used a voltage source of sine wave at 1 V p-p and recorded the output voltage at 15 different frequencies and created the frequency response plot. The table and plot are provided below.

Frequency and Output Voltage table:

frequency	Vout	Gain	Gain _{dB}
100	22 mV	0.022	-33.15
200	18 mV	0.018	-34.84
500	26 mV	0.026	-31.70
600	22 mV	0.022	-33.15
1000	34 mV	0.034	-29.37
10,000	414 mV	0.414	-7.65
20,000	630 mV	0.630	-4.01
30,000	720 mV	0.720	-2.85
40,000	760 mV	0.760	-2.38
50,000	790 mV	0.790	-2.04
60,000	800 mV	0.800	-1.93
70,000	800 mV	0.800	-1.93
80,000	810 mV	0.810	-1.83
90,000	820 mV	0.820	-1.72
100,000	820 mV	0.820	-1.72

Frequency Response Plot:



Conclusion:

We were able to successfully complete this lab. We did run into a few issues as we were completing this lab. One of these issues was in part 1. We originally built the circuit using the components we found by using the correct formulas. We did not get our desired result and we got an output voltage that was much higher than expected. It wasn't until we started to experiment with different potentiometers and different resistors that we were able to get a good result. It was surprising to our group since these new components did not match with our math but somehow were able to give the circuit the correct adjustable gain. Another issue we ran into while completing this lab was that we did not know how to implement the bandpass filter in part 2. We tried looking up examples online, but we were not sure if they were the correct circuit diagrams. We were able to find the correct bandpass circuit diagram on one of our lecture slides. We were more confident with this circuit diagram since it was provided to us by our professor. I think we learned a lot in this lab. We were able to learn more about filters and how to create frequency response plots. It was a challenging but informative lab and I'm sure we will continue to use this information as we continue in the course.